

Lab Goal : This lab was designed to teach you more about graphs, mazes, and recursion.

Lab Description : Oh no! You are trapped in the dreaded sauce trap and must avoid getting lost in the sauce. You were on a school field trip when you got stuck in the trap. Someone took all of the wheels of the bus and hid them throughout the trap, so we must retrieve all of them before we can leave. Field trips can get real interesting. The trap is similar to a maze game you'd play as a child, but much more deadly. Every time you take a step, some of the walls will change, and the maze has 3 "phases" that affect how the walls move. The other deadly twist is that after a certain amount of time has passed, the entire trap will sink into the sauce, and your class will forever be lost in the sauce. There is not just one exit, but there are multiple breaks in the outer wall of the trap through which we can escape. Every second we can take a step up, down, right, or left, but every step means we have one less second left to escape the trap. We begin the trap in phase 1, but every second it shifts, going from 1 to 2, 2 to 3, and 3 back to 1, over and over again until time has elapsed. If we make it to a break in the wall in the wall when time runs out, we are safe ($\leq$, I couldn't come up with a funny way to say this, if your steps is equal to the allowed time you are safe, and count the steps to get to the exit points, don't add an extra step to get out). An exit is not guaranteed to exist, sometimes the trap is inescapable. Backtracking is allowed.

Input : There will be an integer n indicating the number of test cases to follow. Each test case will begin with 3 space separated integers, r , c , and t , denoting the number of rows and columns in the input, and the maximum number of steps allowed to escape, respectively. The next r lines will consist of c characters denoting the shape of the sauce trap, the characters may be any of the following:

- 'S' – This character denotes the starting point, but should be treated as a '1' for escape purposes.
- '1' – This character is a wall in phases 2 and 3, but an open space in phase 1.
- '2' – This character is a wall in phases 1 and 3, but an open space in phase 2.
- '3' – This character is a wall in phases 2 and 1, but an open space in phase 3.
- '#' – This character is a wall in all 3 phases.
- 'o' – This character is one of the wheels of the bus we must retrieve.

Output : If you make it out before the trap sinks into the sauce, then output the following " Cruisin on down Main Street. We made it out in X steps!", where X is the number of steps required to escape. If you don't escape because you run out of time, or can't make it to an exit point, output "We got lost in the sauce. Call us Gucci Mane.". If there were no exit points to begin with, output "The sauce is inevitable.".

Sample Data :

```
10
8 8 20
###2###
#S231o3#
#####2o#
#321322#
#212#o3#
331o####
#32123o#
##3###2#
5 5 19
###S#
#oo2#
#23o#
#o3o#
#2###
6 6 42
#####
##S23#
#o231#
21312#
#oo23#
#3####
4 4 100
####
#So#
#oo#
####
8 8 15
###2####
#S231o3#
#####2o#
#321322#
#212#o3#
331o####
#32123o#
##3###2#
15 15 62
#####1#####
#3#23#2#32o#123
#2So2#1#123#o1#
#323323#o3##223
213112o121231#o
#####31####32
#2132o####o321#
#3###32#132###2
#1#321132o123##
32###31211#####
#32232#3#3212##
#122333#12312##
#232213#####2##
#31o3123123123#
#####2#####1#
```

Files Needed ::

No starter files provided.

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```

15 15 60
#####1#####
#3#23#2#23o#123
#2So2#1#123#o1#
#323123#o3##223
213123o221231#o
#####31####32
#2132o####o321#
#3###32#132###2
#1#321132o123##
32###31211#####
#32232#3#3212##
#122333#12312##
#232213#####2##
#31o3123123123#
#####2#####1#
8 8 212
#####
#####
#####
#####
#####
#####
#####
#####
6 6 20
#####
S23o23
#####1
o32132
2#####
312o##
6 6 10
#####
S23o23
#####1
o32132
2#####
312o##

```

Sample Output :

```

Cruisin on down Main Street. We made it out in 19 steps!
Cruisin on down Main Street. We made it out in 10 steps!
We got lost in the sauce. Call us Gucci Mane.
The sauce is inevitable.
We got lost in the sauce. Call us Gucci Mane.
Cruisin on down Main Street. We made it out in 61 steps!
We got lost in the sauce. Call us Gucci Mane.
The sauce is inevitable.
Cruisin on down Main Street. We made it out in 17 steps!
We got lost in the sauce. Call us Gucci Mane.

```